

Elia Gatti

Master's Student in Computer Science | HPC & GPU Software Engineer

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Work Experience

- Bitmovin - Software Engineer Intern** June 2026-Sept 2026
- Software engineering internship at **Bitmovin**, a leader in video streaming infrastructure (encoding, playback, analytics).
- Dedagroup - Internship Software Developer** May 2024-Sept 2024
- Migrated the 'TEN' treasury application from legacy Windows to Linux servers, **halving VM resource requirements**.
 - Developed reactive **HTMX frontend features** and maintained the Java/Spring backend.

Education

Expected 2026	Master's Degree in Computer Science	University of Trento
Graduated 2024	Bachelor's Degree in Computer Science	University of Trento

Projects

- Sparse Matrix-Vector Multiplication (SpMV) Optimization** 
- Developed a **hybrid-adaptive CUDA kernel** for NVIDIA A30.
 - Profiled execution time and GFLOPS, focusing on memory access coalescing and occupancy tuning.
- Parallel Minimum Spanning Tree (CPU)** 
- Implemented parallel MST algorithms (Borůvka / Kruskal) using OpenMP and MPI.
 - Analyzed speedup and scalability on multi-core CPU clusters.
- Distributed Key-Value Storage System** 
- Built with **Akka actors** – consistent hashing, data replication, fault tolerance.
 - Handled concurrent read/write requests and node failures without client disruption.
- MovieMatch – Recommendation Microservices** 
- Designed modular recommendation backend (Python/FastAPI, Docker Compose).
 - Iteratively refactored** from an over-engineered 15-service architecture into a consolidated, maintainable design.
 - Implemented API gateway and integrated third-party streaming availability providers.
- Guardian — Privacy-First Eldercare** 
- EuroTech-HK Hackathon, Munich, June 2026*
- Built zero-config elderly monitoring with mmWave radar + on-device Gemma 4 LLM — no cameras, no wearables, no cloud.
 - Cross-border eldercare: behavioral drift tracking with real-time crisis alerting.
- DWT-SVD Watermarking (1st Place)** 
- CTM Challenge, Trento, 2025*
- Engineered SVD-based watermark embedding within DWT blocks, robust to compression/noise/transforms.
 - Built automated attack suite (ROC validation).

Skills

Languages	C, C++, CUDA, Java, Python
Frameworks & Tools	FastAPI, Docker, Git, Linux, OpenMP, MPI
Concepts	GPU programming (memory hierarchy, kernel optimization), parallel algorithms, distributed systems, performance benchmarking